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Method for manufacturing semiconductor device

Abstract

A method for manufacturing a semiconductor device, includes forming source and drain electrodes on a semiconductor layer provided above a substrate; forming a first insulating film covering a surface of the semiconductor layer, between the source and drain electrodes, forming a second insulating film on the first insulating film, forming a mask on the second insulating film, the mask having an opening between the source and drain electrodes in a plan view viewed in a direction perpendicular to a substrate surface, forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first and second insulating films through the opening, and forming a gate electrode on the first and second insulating films, the gate electrode making a Schottky contact with the semiconductor layer through the first and second gate openings.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims priority to Japanese Patent Application No. 2022-

TECHNICAL FIELD

(2) The present disclosure relates to methods for manufacturing semiconductor devices.

BACKGROUND

(3) For a high electron mobility transistor (HEMT), there is a proposed structure in which a portion of a gate electrode extending on an insulating film toward a drain electrode is larger than a portion of the gate electrode extending on the insulating film toward a source electrode, in order to reduce electric field concentration. If the portion of the gate electrode extending on the insulating film toward the source electrode were large, an electrostatic capacitance between the gate electrode and a channel (two-dimensional electron gas) would become large.

(4) Related art include Japanese Laid-Open Patent Publication No. 2013-77621, Japanese Laid-Open Patent Publication No. 2016-157983, Japanese Laid-Open Patent Publication No. 2007-88371, U.S. Pat. Nos. 7,501,669, and 7,812,369, for example.

(5) According to related methods for manufacturing semiconductor devices, it is difficult to simultaneously relax the electric field concentration and reduce an increase in the electrostatic capacitance.

SUMMARY

(6) It is one object of the present disclosure to provide a method for manufacturing a semiconductor device capable of simultaneously relaxing of the electric field concentration and reducing an increase in the electrostatic capacitance.

(7) According to one aspect of the present disclosure, a method for manufacturing a semiconductor device, includes forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate; forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, and an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

(8) The object and advantages of the embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

(9) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross sectional view (part 1) illustrating a method for manufacturing a semiconductor device according to a first embodiment;

(2) FIG. 2 is a cross sectional view (part 2) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(3) FIG. 3 is a cross sectional view (part 3) illustrating the method for manufacturing the

semiconductor device according to the first embodiment;

(4) FIG. 4 is a cross sectional view (part 4) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(5) FIG. 5 is a cross sectional view (part 5) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(6) FIG. 6 is a cross sectional view (part 6) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(7) FIG. 7 is a sectional view (part 7) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(8) FIG. 8 is a cross sectional view (part 8) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(9) FIG. 9 is a cross sectional view (part 9) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(10) FIG. 10 is a cross sectional view (part 10) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(11) FIG. 11 is a cross sectional view (part 11) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(12) FIG. 12 is a cross sectional view (part 12) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(13) FIG. 13 is a cross sectional view (part 13) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

(14) FIG. 14 is a diagram generally illustrating a finger gate structure;

(15) FIG. 15 is a cross sectional view (part 1) illustrating the method for manufacturing the semiconductor device according to a second embodiment;

(16) FIG. 16 is a cross sectional view (part 2) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(17) FIG. 17 is a cross sectional view (part 3) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(18) FIG. 18 is a cross sectional view (part 4) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(19) FIG. 19 is a sectional view (part 5) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(20) FIG. 20 is a cross sectional view (part 6) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(21) FIG. 21 is a sectional view (part 7) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(22) FIG. 22 is a cross sectional view (part 8) depicting the method for manufacturing the semiconductor device according to the second embodiment;

(23) FIG. 23 is a cross sectional view (part 9) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(24) FIG. 24 is a cross sectional view (part 10) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(25) FIG. 25 is a cross sectional view (part 11) depicting the method for manufacturing the semiconductor device according to the second embodiment;

(26) FIG. 26 is a cross sectional view (part 12) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(27) FIG. 27 is a cross sectional view (part 13) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

(28) FIG. 28 is a cross sectional view (part 1) illustrating a first example of a mold forming method;

- (29) FIG. **29** is a cross sectional view (part 2) illustrating the first example of the mold forming method;
- (30) FIG. **30** is a cross sectional view (part 3) illustrating the first example of the mold forming method;
- (31) FIG. **31** is a cross sectional view (part 4) illustrating the first example of the mold forming method;
- (32) FIG. **32** is a cross sectional view (part 5) illustrating the first example of the mold forming method;
- (33) FIG. **33** is a cross sectional view (part 6) illustrating the first example of the mold forming method;
- (34) FIG. **34** is a cross sectional view (part 7) illustrating the first example of the mold forming method;
- (35) FIG. **35** is a cross sectional view (part 1) illustrating a second example of the mold forming method;
- (36) FIG. **36** is a cross sectional view (part 2) illustrating the second example of the mold forming method;
- (37) FIG. **37** is a cross sectional view (part 3) illustrating the second example of the mold forming method;
- (38) FIG. **38** is a cross sectional view (part 4) illustrating the second example of the mold forming method; and
- (39) FIG. **39** is a cross sectional view (part 5) illustrating the second example of the mold forming method.

DETAILED DESCRIPTION

(40) A description will hereinafter be given of embodiments of the present invention with reference to the drawings.

(41) [1] A method for manufacturing a semiconductor device according to one aspect of the present disclosure, includes: forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate; forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, and an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

(42) In the opening formed in the mask, the angle formed between the first side surface on the side closer to the source electrode and the upper surface of the substrate, is larger than the angle formed between the second side surface on the side closer to the drain electrode and the upper surface of the substrate. For this reason, the opening of the mask can easily widen on the side closer to the drain electrode than on the side closer to the source electrode during etching. Further, during the etching, the etching rate of the second insulating film is higher than the etching rate of the first insulating film. Hence, the second gate opening formed in the second insulating film is more easily affected by the shape of the opening of the mask than the first gate opening formed in the first insulating film. Accordingly, the portion of the first insulating film exposed from the second

insulating film can easily widen on the side closer to the drain electrode than on the side closer to the source electrode. Thus, the portion of the gate electrode extending on the first insulating film toward the drain electrode can be made larger than the portion of the gate electrode extending on the first insulating film toward the source electrode. As a result, it is possible to simultaneously relax the electric field concentration and reduce the increase in the electrostatic capacitance.

(43) [2] In [1], a distance between the first gate opening and the second gate opening on a side closer to the drain electrode than the first gate is to the drain electrode opening may be longer than a distance between the first gate opening and the second gate opening on a side closer to the source electrode than the first gate opening is to the source electrode. In this case, it is easy to simultaneously relax of the electric field concentration and reduce of an increase in the electrostatic capacitance.

(44) [3] In [1] or [2], an angle between the first side surface and the upper surface may be in a range greater than or equal to 85° and less than or equal to 90° , and an angle between the second side surface and the upper surface may be in a range greater than or equal to 45° and less than or equal to 60° . In this case, particularly during the etching, the opening of the mask can easily widen on the side closer to the drain electrode than on the side closer to the source electrode.

(45) [4] In any one of [1] to [3], the first insulating film may be a first silicon nitride film having a first refractive index, and the second insulating film may be a second silicon nitride film having a second refractive index lower than the first refractive index. In this case, the etching rate of the second insulating film can easily be made higher than the etching rate of the first insulating film.

(46) [5] In any one of [1] to [4], the forming the mask may include forming a positive photosensitive film on the second insulating film, exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film, and the exposing the photosensitive film may include an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface. In this case, it is easy to form the opening with high accuracy.

(47) [6] In any one of [1] to [4], the forming the mask may include forming a positive photosensitive film on the second insulating film, exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film. In this case, the opening can easily be formed with a particularly high accuracy.

(48) [7] In any one of [1] to [4], the forming the mask may include planarizing an upper surface of the second insulating film, forming an uncured film of a resin on the planarized second insulating film, forming a cured film by curing the uncured film while pressing a mold against the uncured film, and removing the mold from the cured film, and the mold may include a base, and a protrusion protruding from the base and having a shape corresponding to a shape of the opening. In this case, the mask can easily be formed with a high accuracy.

(49) [8] In [7], the resin may be an ultraviolet curable resin, and the forming the cured film may include irradiating ultraviolet light on the uncured film through the mold. In this case, the mask can easily be formed with a high accuracy.

(50) [9] In [7] or [8], the source electrode and the drain electrode may extend in a first direction parallel to the upper surface, and be alternately formed at multiple locations in a second direction parallel to the upper surface and perpendicular to the first direction, and the gate electrode may be formed at each location between the source electrode and the drain electrode that are adjacent to each other in the second direction. In this case, a transistor having a finger gate structure can easily be formed.

(51) [10] In any one of [1] to [9], an etching rate of the second insulating film may be four times or more higher than an etching rate of the first insulating film during the etching of the forming the

first gate opening. In this case, the portion of the first insulating film exposed from the second insulating film can easily be made larger on the side closer to the drain electrode than on the side closer to the source electrode.

Details of Embodiments of Present Disclosure

(52) Hereinafter, embodiments of the present disclosure will be described in detail, but the present disclosure is not limited to the described embodiments. In the present specification and drawings, constituent elements having substantially the same functional configuration are denoted by the same reference numerals, a redundant description thereof may be omitted.

First Embodiment

(53) A first embodiment will be described. The first embodiment relates to a method for manufacturing a semiconductor device including a GaN-HEMT having a nitride semiconductor as a main constituent material thereof. FIG. 1 through FIG. 13 are cross sectional views illustrating the method for manufacturing the semiconductor device according to the first embodiment.

(54) First, as illustrated in FIG. 1, a buffer layer 12, an electron transit layer (or channel layer) 14, an electron supply layer (or barrier layer) 16, and a capping layer 18 are successively formed on a substrate 10. The substrate 10 is a silicon carbide (SiC) substrate having an upper surface that is (0001)-oriented, for example. The buffer layer 12 is an aluminum nitride (AlN) layer having a thickness in a range greater than or equal to 5 nm and less than or equal to 100 nm, for example. The electron transit layer 14 is an undoped gallium nitride (GaN) layer having a thickness of approximately 1000 nm, for example. The electron supply layer 16 is an n-type aluminum gallium nitride (AlGaN) layer having a thickness of approximately 20 nm, for example. The capping layer 18 is an n-type GaN layer having a thickness of approximately 5 nm, for example. An n-type impurity used in this embodiment is silicon (Si) or germanium (Ge), for example. A laminating direction of the buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18 is the [0001]-direction, for example. The buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18 are formed by metal organic chemical vapor deposition (MOCVD), for example. A two-dimensional electron gas (2DEG) 72 is present near an upper surface of the electron transit layer 14. A laminated structure formed by the buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18, is an example of a semiconductor layer.

(55) Next, a first insulating film 21 is formed on the capping layer 18. The first insulating film 21 is a silicon nitride (SiN) film, for example. The first insulating film 21 has a thickness in a range greater than or equal to 10 nm and less than or equal to 20 nm, for example. The silicon nitride film (an example of a first silicon nitride film) used for the first insulating film 21 is preferably has a composition represented by a formula $\text{Si}_{2.0}\text{N}_{3.0}$. A refractive index of the $\text{Si}_{2.0}\text{N}_{3.0}$ is 2.0. The first insulating film 21 can be formed by chemical vapor deposition (CVD), for example.

(56) Next, as illustrated in FIG. 2, a resist mask 80 is formed on the first insulating film 21. The resist mask 80 has an opening 81 that exposes a portion of the first insulating film 21, and an opening 82 that exposes another portion of the first insulating film 21.

(57) Next, as illustrated in FIG. 3, openings 31 and 41 are formed in the first insulating film 21, the capping layer 18, and the electron supply layer 16 by reactive ion etching (RIE). The opening 31 is communicates to the opening 81, and the opening 41 communicates to the opening 82. A reactive gas including fluorine (F) may be used to etch the first insulating film 21. A reactive gas including chlorine (Cl) may be used to etch the capping layer 18 and the electron supply layer 16. The openings 31 and 41 are formed to a depth such that a portion of the electron supply layer 16 remains, for example.

(58) Next, as illustrated in FIG. 4, a source electrode 32 is formed in the opening 31, and a drain electrode 42 is formed in the opening 41. When forming the source electrode 32 and the drain electrode 42, a metal film is formed by vacuum deposition in a state where the resist mask 80 remains, and the resist mask 80 is removed after forming the metal film. Although the metal film is

formed not only inside the openings **31** and **41** but also on the resist mask **80**, the metal film on the resist mask **80** is removed when the resist mask **80** is removed. In other words, lift-off of the metal film together with the resist mask **80** is performed. The resist mask **80** can be removed using an organic solvent, for example. An alloying heat treatment is performed after the resist mask **80** is removed. The source electrode **32** and the drain electrode **42** include a Ta film and an Al film, for example. The source electrode **32** and the drain electrode **42** make ohmic contact with the 2DEG **72**.

(59) Next, as illustrated in FIG. 5, A second insulating film **22** is formed on the first insulating film **21**, the source electrode **32**, and the drain electrode **42**. The second insulating film **22** is a silicon nitride (SiN) film, for example. The second insulating film **22** may be thicker than the first insulating film **21**. The second insulating film **22** has a thickness in a range greater than or equal to 30 nm and less than or equal to 40 nm, for example. A density of the second insulating film **22** is preferably lower than a density of the first insulating film **21**. In addition, a refractive index (an example of a second refractive index) of the second insulating film **22** is preferably lower than the refractive index (an example of a first refractive index) of the first insulating film **21**. The refractive index of the silicon nitride film (an example of a second silicon nitride film) used for the second insulating film **22** is 1.85, for example. The second insulating film **22** can be formed by CVD, for example.

(60) Next, as illustrated in FIG. 6, a positive photoresist film **101** is formed on the second insulating film **22**. The photoresist film **101** is an example of a photosensitive film.

(61) Next, as illustrated in FIG. 7, a photosensitive region **121** is formed, by exposure, in a portion of the photoresist film **101** between the source electrode **32** and the drain electrode **42**. The exposure is performed by reduction exposure, for example. When performing the exposure, light **150** is irradiated from a direction inclined from a direction perpendicular to an upper surface **10A** of the substrate **10** toward the drain electrode **42**. The photosensitive region **121** has side surfaces **151** and **152** that are parallel to each other in a cross sectional view thereof. The side surface **151** is closer to the source electrode **32** than the side surface **152** is to the source electrode **32**. An angle formed by the upper surface **10A** and the side surfaces **151** and **152** may be in a range greater than or equal to 45° and less than or equal to 60°, for example. The photosensitive region **121** has a cross section having a parallelogram shape.

(62) Next, as illustrated in FIG. 8, a photosensitive region **122** partially overlapping the photosensitive region **121** is formed, by exposure, in a portion of the photoresist film **101** between the source electrode **32** and the drain electrode **42**. The exposure is performed by reduction exposure, for example. When performing the exposure, light **160** is irradiated from the direction perpendicular to the upper surface **10A** of the substrate **10**. The photosensitive region **122** is formed such that the side surface **152** remains, but the side surface **151** is incorporated into the photosensitive region **122**. The photosensitive region **122** has a side surface **161** located at a position closer to the source electrode **32**, than the position where the side surface **151** existed is to the source electrode **32**. Although the upper surface **10A** and the side surface **161** are perpendicular to each other, the side surface **161** may be slightly inclined from the direction perpendicular to the upper surface **10A** of the substrate **10** toward the source electrode **32** or the drain electrode **42**. An angle formed by the upper surface **10A** and the side surface **161** is in a range greater than or equal to 85° and less than or equal to 90°, for example. The photosensitive region **122** has a cross section having a rectangular shape.

(63) A photosensitive region **120** is formed from the photosensitive regions **121** and **122** partially overlapping each other. The photosensitive region **120** has the side surfaces **152** and **161**. The photosensitive region **120** has a cross section having a trapezoidal shape with two adjacent apexes forming an angle of 90°, where one of the two adjacent apexes is formed by the side surface **161** and an upper surface of the photosensitive region **122**, and the other of the two adjacent apexes is formed by the side surface **161** and a lower surface of the photosensitive region **122**.

(64) Next, as illustrated in FIG. 9, the photosensitive region **120** is removed by developing the photoresist film **101**. As a result, an opening **110** having side surfaces **112** and **114** is formed in the photoresist film **101**. The side surface **112** is formed at the position of the side surface **161** of the photosensitive region **120**, and the side surface **114** is formed at the position of the side surface **152** of the photosensitive region **120**. Accordingly, the side surface **112** is closer to the source electrode **32** than the side surface **114** is to the source electrode **32**. In the opening **110**, an angle formed by the side surface **112** on the side of the source electrode **32** and the upper surface **10A** of the substrate **10** is larger than an angle formed by the side surface **114** on the side of the drain electrode **42** and the upper surface **10A** of the substrate **10**. The angle formed by the upper surface **10A** and the side surface **112** may be in a range greater than or equal to 85° and less than or equal to 90° for example, and the angle formed by the upper surface **10A** and the side surface **114** may be in a range greater than or equal to 45° and less than or equal to 60° , for example. An alkaline developer is used to develop the photoresist film **101**, for example. The photoresist film **101** formed with the opening **110** is an example of a mask.

(65) Next, as illustrated in FIG. 10, the first gate opening **50** is formed in the first insulating film **21**, and the second gate opening **60** is formed in the second insulating film **22**, by etching the first insulating film **21** and the second insulating film **22** through the opening **110**. RIE is used for this etching, for example. The first gate opening **50** has side surfaces **52** and **54** in the cross sectional view. The side surface **52** is located at a position closer to the source electrode **32** than the side surface **54** is to the source electrode **32**. The second gate opening **60** has side surfaces **62** and **64** in the cross sectional view. The side surface **62** is located at a position closer to the source electrode **32** than the side surface **64** is to the source electrode **32**. A reactive gas including fluorine (F) may be used to etch the first insulating film **21** and the second insulating film **22**. This etching is performed under a condition that the etching rate of the second insulating film **22** becomes higher than the etching rate of the first insulating film **21**. For example, the etching rate of the second insulating film **22** is set to approximately four times the etching rate of the first insulating film **21**. For example, the etching rate of the first insulating film **21** is in a range greater than or equal to 1 nm/minute and less than or equal to 3 nm/minute, and the etching rate of the second insulating film **22** is in a range greater than or equal to 4 nm/minute and less than or equal to 12 nm/minute.

(66) During this etching, the opening **110** of the photoresist film **101** widens while maintaining the shape of the side surfaces **112** and **114**. In this state, the side surface **112** is perpendicular to the upper surface **10A**, but the side surface **114** is inclined from the side surface **112**. For this reason, the side surface **114** is more easily etched than the side surface **112**, and a moving amount of the side surface **114** is larger than a moving amount of the side surface **112**. In addition, the second gate opening **60** of the second insulating film **22** is formed following the shape of the opening **110**. Accordingly, the side surface **62** of the second gate opening **60** becomes continuous with the side surface **112**, and the side surface **64** of the second gate opening **60** becomes continuous with the side surface **114**. The angle formed by the upper surface **10A** and the side surface **62** may be in a range greater than or equal to 85° and less than or equal to 90° , for example, and the angle formed by the upper surface **10A** and the side surface **64** may be in a range greater than or equal to 45° and less than or equal to 60° , for example.

(67) On the other hand, because the first insulating film **21** is etched at a rate that is approximately 1/4 that of the second insulating film **22**, even when the opening **110** is widened, the first gate opening **50** is hardly affected by the widening of the opening **110**. Accordingly, the first gate opening **50** is mainly formed below the portion exposed inside the opening **110** at the start of etching the second insulating film **22**. Hence, the side surface **52** of first gate opening **50** becomes discontinuous with the side surface **62**, and side surface **54** of first gate opening **50** becomes discontinuous with the side surface **64**. The side surfaces **52** and **54** may be a surface perpendicular to the upper surface **10A**, or may be a surface inclined from the surface perpendicular to the upper surface **10A**.

(68) Because the first gate opening **50** and the second gate opening **60** are formed by such a mechanism, a distance L_s between the side surface **52** of the first gate opening **50** and the side surface **62** of the second gate opening **60** becomes smaller than a distance L_d between the side surface **54** of the first gate opening **50** and the side surface **64** of the second gate opening **60**.

(69) After forming the first gate opening **50** and the second gate opening **60**, the photoresist film **101** is removed as illustrated in FIG. **11**. The photoresist film **101** can be removed by using an organic solvent, oxygen plasma, or the like, for example.

(70) Next, as illustrated in FIG. **12**, a gate electrode **71** is formed on the first insulating film **21** and the second insulating film **22**. The gate electrode **71** can be formed by vapor deposition and lift-off, for example. The gate electrode **71** includes a Ni film and a Au film, for example. The gate electrode **71** makes a Schottky contact with the capping layer **18**, through the first gate opening **50** and the second gate opening **60**.

(71) Next, as illustrated in FIG. **13**, a third insulating film **23** covering the gate electrode **71**, the source electrode **32**, and the drain electrode **42**, is formed. The third insulating film **23** is a silicon nitride (SiN) film, for example. The silicon nitride film can be formed by CVD, for example. Next, an opening **33** that exposes a portion of the source electrode **32**, and an opening **43** that exposes a portion of the drain electrode **42**, are formed in the third insulating film **23** and the second insulating film **22**. The openings **33** and **43** are formed by RIE using a resist mask (not illustrated), for example.

(72) Thereafter, interconnects or the like are formed, as required. Hence, the semiconductor device **100** including the GaN-HEMT can be manufactured in the above described manner.

(73) According to the first embodiment, in the opening **110** formed in the photoresist film **101**, the angle formed by the side surface **112** on the side of the source electrode **32** and the upper surface **10A** of the substrate **10** is larger than the angle formed by the side surface **114** on the side of the drain electrode **42** and the upper surface **10A** of the substrate **10**. For this reason, as described above, the opening **110** easily widens on the side of the drain electrode **42** during the etching than on the side of the source electrode **32**. In addition, during the etching, the etching rate of the second insulating film **22** is higher than the etching rate of the first insulating film **21**. Hence, the second gate opening **60** is more easily affected by the shape of the opening **110** than the first gate opening **50**. Thus, the portion of the first insulating film **21** exposed from the second insulating film **22** easily widens on the side of the drain electrode **42** than on the side of the source electrode **32**. For this reason, the portion of the gate electrode **71** widening on the first insulating film **21** toward the drain electrode **42** can be made larger than the portion of the gate electrode **71** widening on the first insulating film **21** toward the source electrode **32**. As a result, it is possible to simultaneously relax the electric field concentration and reduce the increase in electrostatic capacitance.

(74) Moreover, when forming the opening **110**, two exposures with different light irradiating directions are performed, and thus, the opening **110** can be easily be formed with a high accuracy. The photosensitive region **120** may be formed by a single exposure using a shifted telecentric optical system, instead of performing the two exposures. It is also possible in this case to easily form the opening **110** with a high accuracy.

(75) When the angle formed by the upper surface **10A** and the side surface **112** is in a range greater than or equal to 85° and less than or equal to 90° , and the angle formed by the upper surface **10A** and the side surface **114** is in a range greater than or equal to 45° and less than or equal to 60° , the opening **110** in particular easily widens on the side of the drain electrode **42** than on the side of the source electrode **32** during etching. The angle formed by the upper surface **10A** and the side surface **112** is more preferably in a range greater than or equal to 87° and less than or equal to 90° , and the angle formed by the upper surface **10A** and the side surface **114** is more preferably in a range greater than or equal to 45° and less than or equal to 55° .

(76) In the case where the first insulating film **21** is a first silicon nitride film having a first refractive index (for example, 2.00), and the second insulating film **22** is a second silicon nitride

film having a second refractive index (for example, 1.85) lower than the first refractive index, the etching rate of the second insulating film **22** can easily be made higher than the etching rate of the first insulating film **21**. The refractive index of the silicon nitride film can be measured using a spectroscopic ellipsometer or a spectroscopic reflectometry film thickness measuring device. Further, the silicon nitride film has wavelength dispersion. The refractive index of the silicon nitride film in the present disclosure is defined as the refractive index at a wavelength of 632.8 nm (HeNe laser).

(77) In the case where the etching rate of the second insulating film **22** is four times or more higher than the etching rate of the first insulating film **21**, the portion of the first insulating film **21** exposed from the second insulating film **22** can easily be widened on the side of the drain electrode **42** than on the side of the source electrode **32**. More preferably, the etching rate of the second insulating film **22** is five times or more higher than the etching rate of the first insulating film **21**.

Second Embodiment

(78) Next, a second embodiment will be described. The second embodiment relates to a method for manufacturing a semiconductor device including a GaN-HEMT having a nitride semiconductor as a main constituent material thereof. The second embodiment is particularly suitable for a GaN-HEMT having a finger gate structure. FIG. **14** is a diagram generally illustrating the finger gate structure. FIG. **15** through FIG. **27** are cross sectional views illustrating the method for manufacturing the semiconductor device according to the second embodiment.

(79) In the finger gate structure, as illustrated in FIG. **14**, the gate electrode **71**, the source electrode **32**, and the drain electrode **42** extend in a first direction parallel to the upper surface of the substrate. In addition, a plurality of source electrodes **32** and drain electrodes **42** are alternately formed in a second direction that is parallel to the upper surface of the substrate and is perpendicular to the first direction. One gate electrode **71** is formed between each pair of the source electrode **32** and the drain electrode **42** that are adjacent to each other in the second direction.

(80) First, in the second embodiment, the buffer layer **12**, the electron transit layer **14**, the electron supply layer **16**, and the capping layer **18** are successively formed on the substrate **10**, as illustrated in FIG. **15**, similar to the first embodiment. Next, the first insulating film **21** is formed on the capping layer **18**.

(81) Next, as illustrated in FIG. **16**, a resist mask **80** is formed on the first insulating film **21**. The resist mask **80** has a plurality of openings **81**, and a plurality of openings **82**. The openings **81** and the openings **82** are alternately arranged in the second direction.

(82) Next, as illustrated in FIG. **17**, the openings **31** and **41** are formed in the first insulating film **21**, the capping layer **18**, and the electron supply layer **16** by RIE, similar to the first embodiment. In this state, the openings **31** and **41** are formed to extend in the first direction parallel to the upper surface **10A** of the substrate **10**. In addition, the plurality of openings **31** and **41** are alternately formed in the second direction that is parallel to the upper surface **10A** and is perpendicular to the first direction.

(83) Next, as illustrated in FIG. **18**, the source electrode **32** is formed inside the opening **31**, and the drain electrode **42** is formed inside the opening **41**, similar to the first embodiment. In this state, the source electrode **32** and the drain electrode **42** are formed to extend in the first direction. In addition, the plurality of source electrodes **32** and the plurality of drain electrodes **42** are alternately formed in the second direction.

(84) Next, as illustrated in FIG. **19**, the second insulating film **22** is formed on the first insulating film **21**, the source electrode **32**, and the drain electrode **42**, similar to the first embodiment.

(85) Next, as illustrated in FIG. **20**, an upper surface of the second insulating film **22** is planarized. The upper surface of the second insulating film **22**, an upper surface of the source electrode **32**, and an upper surface of the drain electrode **42** may coincide with one another. For example, the second insulating film **22** is polished until the upper surface of the source electrode **32** and the upper surface of the drain electrode **42** become exposed. For example, chemical mechanical polishing

(CMP) may be performed as the polishing of the second insulating film **22**. The source electrode **32** and the drain electrode **42** may also be polished.

(86) Next, as illustrated in FIG. **21**, an uncured film **201A** of an ultraviolet (UV) curable resin is formed on the second insulating film **22**.

(87) Further, as illustrated in FIG. **22**, a mold **230** that transmits ultraviolet light is prepared. The mold **230** includes a flat base **235**, and a plurality of protrusions **236** protruding from the base **235**. The protrusions **236** protrude from one surface **235A** of the base **235**, that is, from a lower surface of the base **235**, for example. The protrusions **236** extend in the first direction, and are arranged side by side in the second direction. The protrusions **236** are provided to form openings corresponding to the openings **110** of the first embodiment, in a mask to be formed from the uncured film **201A**. The protrusion **236** has a surface **233** parallel to the surface **235A**, a side surface **232**, and a side surface **234**. The side surface **232** is perpendicular to the surfaces **233** and **235A**, and joins the surface **233** and the surface **235A**. The side surface **234** is inclined from the side surface **232**, and joins the surface **233** and the surface **235A**. The side surfaces **232** and **234** are alternately arranged in the second direction for each of the protrusions **236**. There are two kinds of inclination directions of the side surface **234**. The mold **230** is a quartz mold, for example. A method for forming the mold **230** will be described later.

(88) After forming the uncured film **201A**, the mold **230** is pressed against the uncured film **201A**, as illustrated in FIG. **23**. Then, ultraviolet light **170** is irradiated on the uncured film **201A** through the mold **230**. As a result, a cured film **201** of the UV curable resin is formed. The cured film **201** of the UV curable resin has an opening **210** with side surfaces **212** and **214**. The side surface **212** is formed at the position of the side surface **232** of the protrusion **236**, and the side surface **214** is formed at the position of the side surface **234** of the protrusion **236**. The side surface **212** is located at a position closer to the source electrode **32** than the side surface **214** is to the source electrode **32**. In the opening **210**, an angle formed by the side surface **212** on the side of the source electrode **32** and the upper surface **10A** of the substrate **10** is larger than an angle formed by the side surface **214** on the side of the drain electrode **42** and the upper surface **10A** of the substrate **10**. The angle formed by the upper surface **10A** and the side surface **212** may be in a range greater than or equal to 85° and less than or equal to 90° , for example, and the angle formed by the upper surface **10A** and the side surface **214** may be in a range greater than or equal to 45° and less than or equal to 60° , for example.

(89) Next, as illustrated in FIG. **24**, the mold **230** is removed. The cured film **201** of the UV curable resin, after the mold **230** is removed, is an example of the mask. A residual film treatment (or removal) is preferably performed after removing the mold **230**. The protrusions **236** and the second insulating film **22** may inevitably not make close contact with one another, thereby causing a residual UV curable resin to remain at a bottom of the opening **210**. But even if the residual UV curable resin is present, the residual UV curable resin can be removed by the residual film treatment. The residual film treatment irradiates oxygen plasma, for example.

(90) Next, as illustrated in FIG. **25**, the second gate opening **60** is formed in the second insulating film **22**, and the first gate opening **50** is formed in the first insulating film **21** by RIE, similar to the first embodiment. The distance between the side surface **52** of the first gate opening **50** and the side surface **62** of the second gate opening **60** becomes smaller than the distance between the side surface **54** of the first gate opening **50** and the side surface **64** of the second gate opening **60**, similar to the first embodiment.

(91) Next, as illustrated in FIG. **26**, the cured film **201** of the UV curable resin is removed. The cured film **201** of the UV curable resin can be removed using an organic solvent, oxygen plasma, or the like, for example.

(92) Next, as illustrated in FIG. **27**, the gate electrode **71** is formed on the first insulating film **21** and the second insulating film **22**, the third insulating film **23** is formed to cover the gate electrode **71**, the source electrode **32**, and the drain electrode **42**, and the openings **33** and **43** are formed in

the third insulating film **23**, similar to the first embodiment.

(93) Thereafter, interconnections or the like are formed, as required. Hence, the semiconductor device **200** including the GaN-HEMT can be manufactured in the above described manner.

(94) According to the second embodiment, it is possible to simultaneously relax the electric field concentration and reduce the increase in the electrostatic capacitance, similar to the first embodiment. In addition, because the UV light is irradiated while the mold **230** is pressed against the uncured film **201A** of the UV curable resin, the opening **210** can be easily formed with a high accuracy.

(95) The semiconductor device **200** including the GaN-HEMT having the finger gate structure can also be manufactured by performing three exposures with respect to the photoresist film **101**, in a modification of the first embodiment. That is, the semiconductor device **200** can be manufactured by adding one exposure for forming the side surface **214** having a different inclination direction.

(96) On the other hand, in the second embodiment, the opening **210** can be formed by a single irradiation of the UV light, using the mold **230**. Hence, according to the second embodiment, the GaN-HEMT having the finger gate structure can be easily formed by a small number of processes or steps. The method for forming the opening **210** as in the second embodiment may sometimes be referred to as a UV nanoimprint lithography method.

(97) Next, an example of a method for forming the mold **230** will be described. FIG. **28** through FIG. **34** are cross sectional views illustrating a first example of the method for forming the mold **230**. In this example, the mold **230** is formed from quartz.

(98) First, as illustrated in FIG. **28**, a flat quartz plate **251** is prepared, and a positive photoresist film **252** is formed on one surface **251A** of the quartz plate **251**.

(99) Next, as illustrated in FIG. **29**, a photosensitive region **261** is formed in the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **270** is irradiated from a direction inclined from a direction perpendicular to a surface **251A** of the quartz plate **251**. The photosensitive region **261** has side surfaces **271** and **272** that are parallel to each other in the cross sectional view. As described above, there are two kinds of inclination directions of the side surface **234** of the protrusion **236**. The side surface **271** is formed to correspond to the side surface **234** having one of the two kinds of inclinations. The side surface **272** is formed at a position closer to the side surface **271** than the side surface **234** having the other of the two kinds of inclinations is to the side surface **271**.

(100) Next, as illustrated in FIG. **30**, a photosensitive region **262** partially overlapping the photosensitive region **261** is formed in the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **280** is irradiated from a direction inclined from the direction perpendicular to the surface **251A** of the quartz plate **251** in a direction opposite to the direction in which the light **270** is inclined. The photosensitive region **262** is formed such that the side surface **271** remains, but the side surface **272** is incorporated into the photosensitive region **262**. The photosensitive region **262** has a side surface **281** located at a position on an outer side of the photosensitive region **261** than the position where the side surface **272** existed. The side surface **281** is formed so as to correspond to the side surface **234** having the other of the two kinds of inclinations.

(101) A photosensitive region **260** is formed from the photosensitive regions **261** and **262** partially overlapping each other. The photosensitive region **260** has side surfaces **271** and **281**. The photosensitive region **260** has a cross section having an isosceles trapezoid shape.

(102) Next, as illustrated in FIG. **31**, a photosensitive region **265** is formed in a portion between mutually adjacent photosensitive regions **260** of the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **290** is irradiated from a direction perpendicular to the surface **251A** of the quartz plate **251**. The photosensitive region **265** has side surfaces **291** and **292** that are parallel to each other in the cross sectional view. The side surfaces **291** and **292** are perpendicular to surface **251A**. The side surfaces

291 and **292** are formed to correspond to the side surfaces **232** of the protrusions **236**. For example, the side surface **291** is formed so that an unexposed region of the photoresist film **252** remains between the side surface and the side surface **271**, and the side surface **292** is formed so that an unexposed region of the photoresist film **252** remains between the side surface **281** and the side surface **292**. The photosensitive region **265** has a cross section having a rectangular shape.

(103) Next, as illustrated in FIG. **32**, the photoresist film **252** is developed, to remove the photosensitive regions **260** and **265**. As a result, an opening **243** having side surfaces **241** and **242**, and an opening **248** having side surfaces **246** and **247**, are formed in the photoresist film **252**. The side surface **241** is formed at the position of the side surface **281** of the photosensitive region **260**, and the side surface **242** is formed at the position of the side surface **271** of the photosensitive region **260**. The side surface **246** is formed at the position of the side surface **291** of the photosensitive region **265**, and the side surface **247** is formed at the position of the side surface **292** of the photosensitive region **265**. The angle formed by the surface **251A** and the side surfaces **241** and **242** may be in a range greater than or equal to 45° and less than or equal to 60° , for example, and the angle formed by the surface **251A** and the side surfaces **246** and **247** may be in a range greater than or equal to 85° and less than or equal to 90° , for example. An alkaline developer is used to develop the photoresist film **252**, for example.

(104) Next, as illustrated in FIG. **33**, the base **235** and the plurality of protrusions **236** are formed on the quartz plate **251**, by etching the quartz plate **251** through the openings **243** and **248**. The protrusion **236** has the side surface **232** continuous with the side surface **246** or **247**, and the side surface **234** continuous with the side surface **241** or **242**. RIE is used for the etching described above, for example.

(105) Next, as illustrated in FIG. **34**, the photoresist film **252** is removed. The photoresist film **252** can be removed by using an organic solvent, oxygen plasma, or the like, for example. Hence, the mold **230** made of quartz can be formed in the above described manner.

(106) Next, a second example of the method for forming the mold **230** will be described. FIG. **35** through FIG. **39** are cross sectional views illustrating the second example of the method for forming the mold **230**. In this example, the mold **230** is formed of a resin.

(107) First, as illustrated in FIG. **35**, a master mold **330** made of quartz is formed. The master mold **330** includes a flat base **335**, and a plurality of protrusions **336** protruding from the base **335**. The protrusions **336** protrude from one surface **335A** of the base **335**. The base **335** and the protrusions **336** have shapes similar to the shapes of the base **235** and the protrusions **236**, respectively. The protrusion **336** has a surface **333** parallel to the surface **335A**, a side surface **332**, and a side surface **334**. The master mold **330** has a shape corresponding to a repeating unit including two protrusions **236** that are adjacent to each other in the second direction of the mold **230**.

(108) Next, as illustrated in FIG. **36**, a transfer member **340** is formed on the master mold **330**. The transfer member **340** is a nickel (Ni) film, for example. The transfer member **340** can be formed by nickel electroforming. The transfer member **340** is formed with a surface **341** having concavo-convexes conforming to the protrusions **336** of the master mold **330**.

(109) Next, as illustrated in FIG. **37**, the transfer member **340** is released or parted from the master mold **330**.

(110) Next, as illustrated in FIG. **38**, an uncured resin film **237** is prepared, and the transfer member **340** is pressed against one surface **237A** of the resin film **237**. As a result, concavo-convexes conforming to the surface **341** of the transfer member **340** are formed on the surface **237A**. That is, the concavo-convexes conforming to the protrusions **336** of the master mold **330** are transferred to the surface **237A**. The resin film **237** is a thermoplastic resin film or a UV curable resin film, for example. The resin film **237** may include a fluororesin, for example.

(111) Thereafter, as illustrated in FIG. **39**, the transfer of the concavo-convexes conforming to the protrusions **336** of the master mold **330** using the transfer member **340** is repeated. Then, after all of the concavo-convexes that become the protrusions **236** are formed, the resin film **237** is cured.

The resin film 237 is heated in the case where the resin film 237 is the thermoplastic resin film, and UV light irradiation is performed in the case where the resin film 237 is the UV curable resin film. Hence, the resin mold 230 can be formed in the above described manner.

(112) According to the present disclosure, it is possible to simultaneously relax the electric field concentration and reduce an increase in the electrostatic capacitance.

(113) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

(114) Although the embodiment has been described in detail above, the embodiment is not limited to a specific embodiment, and various modifications and changes can be made within the scope described in the claims.

Claims

1. A method for manufacturing a semiconductor device, comprising: forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate; forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein: the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening, and a distance between the first gate opening and the second gate opening on a side closer to the drain electrode than the first gate opening is to the drain electrode is longer than a distance between the first gate opening and the second gate opening on a side closer to the source electrode than the first gate opening is to the source electrode.

2. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the first insulating film is a first silicon nitride film having a first refractive index, and the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

3. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film, exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film, and the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

4. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film.

5. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the forming the mask includes planarizing an upper surface of the second insulating film, forming an uncured film of a resin on the planarized second insulating film, forming a cured film by curing the uncured film while pressing a mold against the uncured film, and removing the mold from the cured film, and the mold includes a base, and a protrusion protruding from the base and having a shape corresponding to a shape of the opening.

6. The method for manufacturing a semiconductor device as claimed in claim 5, wherein the resin is an ultraviolet curable resin, and the forming the cured film includes irradiating ultraviolet light on the uncured film through the mold.

7. The method for manufacturing the semiconductor device as claimed in claim 5, wherein the source electrode and the drain electrode extend in a first direction parallel to the upper surface, and are alternately formed at multiple locations in a second direction parallel to the upper surface and perpendicular to the first direction, and the gate electrode is formed at each location between the source electrode and the drain electrode that are adjacent to each other in the second direction.

8. A method for manufacturing a semiconductor device, comprising: forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate; forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening, an angle between the first side surface and the upper surface is in a range greater than or equal to 85° and less than or equal to 90° , and an angle between the second side surface and the upper surface is in a range greater than or equal to 45° and less than or equal to 60° .

9. The method for manufacturing the semiconductor device as claimed in claim 8, wherein the first insulating film is a first silicon nitride film having a first refractive index, and the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

10. The method for manufacturing the semiconductor device as claimed in claim 8, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film, exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film, and the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

11. The method for manufacturing the semiconductor device as claimed in claim 8, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film, exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be

formed, and removing the photosensitive region by developing the photosensitive film.

12. A method for manufacturing a semiconductor device, comprising: forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode: forming a second insulating film on the first insulating film: forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate: forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein: the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening, and an etching rate of the second insulating film is four times or more higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

13. The method for manufacturing the semiconductor device as claimed in claim 12, wherein the first insulating film is a first silicon nitride film having a first refractive index, and the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

14. The method for manufacturing the semiconductor device as claimed in claim 12, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film, exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film, and the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

15. The method for manufacturing the semiconductor device as claimed in claim 12, wherein the forming the mask includes forming a positive photosensitive film on the second insulating film, exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film.
